

Nr 10.

ex 1:

$x_4 x_5$	00	01	11	10
00	0	0	-	$\bar{x}_3$
01	x_1	1	-	1
11	0	-	1	x_3
10	-	$\bar{x}_2$	1	0

Pas 1:

$x_4 x_5$	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	0	-	1	0
10	-	0	1	0

$x_4 x_8, x_5 x_6 x_7, x_5 x_6 x_8$

Pas 2:

$x_4 x_5$	00	01	11	10
00	0	0	-	$\bar{x}_3$
01	$\bar{x}_4$	*	-	*
11	0	-	*	x_3
10	-	$\bar{x}_2$	*	0

S-au notat diferite combinațiile indiferente pentru a pune în evidență originea lor diferită. Cu toate acestea ele vor fi luate în considerare în același mod.

$x_4 x_8, x_5 x_8, x_4 x_7, x_5 x_6$

Pas 3: $\bar{x}_4 \cdot x_6 x_8, x_3 \cdot x_5 x_8, \bar{x}_2 \cdot x_6 x_7, x_1 \cdot x_5 x_6$

Pas 4: $f(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8) = x_4 x_8 + x_5 x_6 x_7 + x_5 x_6 x_8 + \bar{x}_4 x_6 x_8 + x_3 x_5 x_8 + \bar{x}_2 x_6 x_7 + x_1 x_5 x_6$

ex 2: FCD

$$f(x_1, x_2, x_3, x_4, x_5) = \sum (1, 3, 7, 8, 13, 16, 27)$$

a) FCD

$$f(x_1, x_2, x_3, x_4, x_5) = \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 x_5 + \bar{x}_1 \bar{x}_2 \bar{x}_3 x_4 x_5 + \bar{x}_1 \bar{x}_2 x_3 x_4 x_5 + \bar{x}_1 x_2 \bar{x}_3 \bar{x}_4 \bar{x}_5 + \bar{x}_1 x_2 x_3 x_4 \bar{x}_5 + \bar{x}_1 x_2 \bar{x}_3 \bar{x}_4 x_5 + x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 \bar{x}_5 + x_1 \bar{x}_2 \bar{x}_3 x_4 x_5 =$$

$$= \bar{x}_1 (\bar{x}_2 \bar{x}_3 \bar{x}_4 x_5) + \bar{x}_1 (\bar{x}_2 \bar{x}_3 x_4 x_5) + \bar{x}_1 (\bar{x}_2 x_3 x_4 x_5) + \bar{x}_1 (x_2 \bar{x}_3 \bar{x}_4 \bar{x}_5) + \bar{x}_1 (x_2 x_3 x_4 \bar{x}_5) + x_1 (\bar{x}_2 \bar{x}_3 \bar{x}_4 \bar{x}_5) + x_1 (\bar{x}_2 \bar{x}_3 x_4 x_5) =$$

$$= \bar{x}_1 \bar{p}_1' + \bar{x}_1 \cdot \bar{p}_3' + \bar{x}_1 \cdot \bar{p}_4' + \bar{x}_1 \cdot \bar{p}_8' + \bar{x}_1 \cdot \bar{p}_{13}' + x_1 \cdot \bar{p}_0' + x_1 \cdot \bar{p}_{11}' =$$

$$= \bar{x}_1 \bar{p}_1' \cdot \bar{x}_1 \cdot \bar{p}_3' \cdot \bar{x}_1 \cdot \bar{p}_4' + \bar{x}_1 \cdot \bar{p}_8' + \bar{x}_1 \cdot \bar{p}_{13}' + x_1 \cdot \bar{p}_0' + x_1 \cdot \bar{p}_{11}' =$$

$$= (x_1 + \bar{p}_1') (x_1 + \bar{p}_3') (x_1 + \bar{p}_4') (x_1 + \bar{p}_8') (x_1 + \bar{p}_{13}') (\bar{x}_1 + \bar{p}_0') (\bar{x}_1 + \bar{p}_{11}') =$$

$$= (x_1 + \bar{p}_1' \bar{p}_3') (x_1 + \bar{p}_4' \bar{p}_8') (x_1 + \bar{p}_{13}') (\bar{x}_1 + \bar{p}_0' \bar{p}_{11}') =$$

$$= (x_1 + \bar{p}_1' \bar{p}_3') (\bar{x}_1 + \bar{p}_0' \bar{p}_{11}') (x_1 + x_1 \bar{p}_{13}' + x_1 \bar{p}_4' \bar{p}_8' + \bar{p}_4' \bar{p}_8' \bar{p}_{13}') =$$

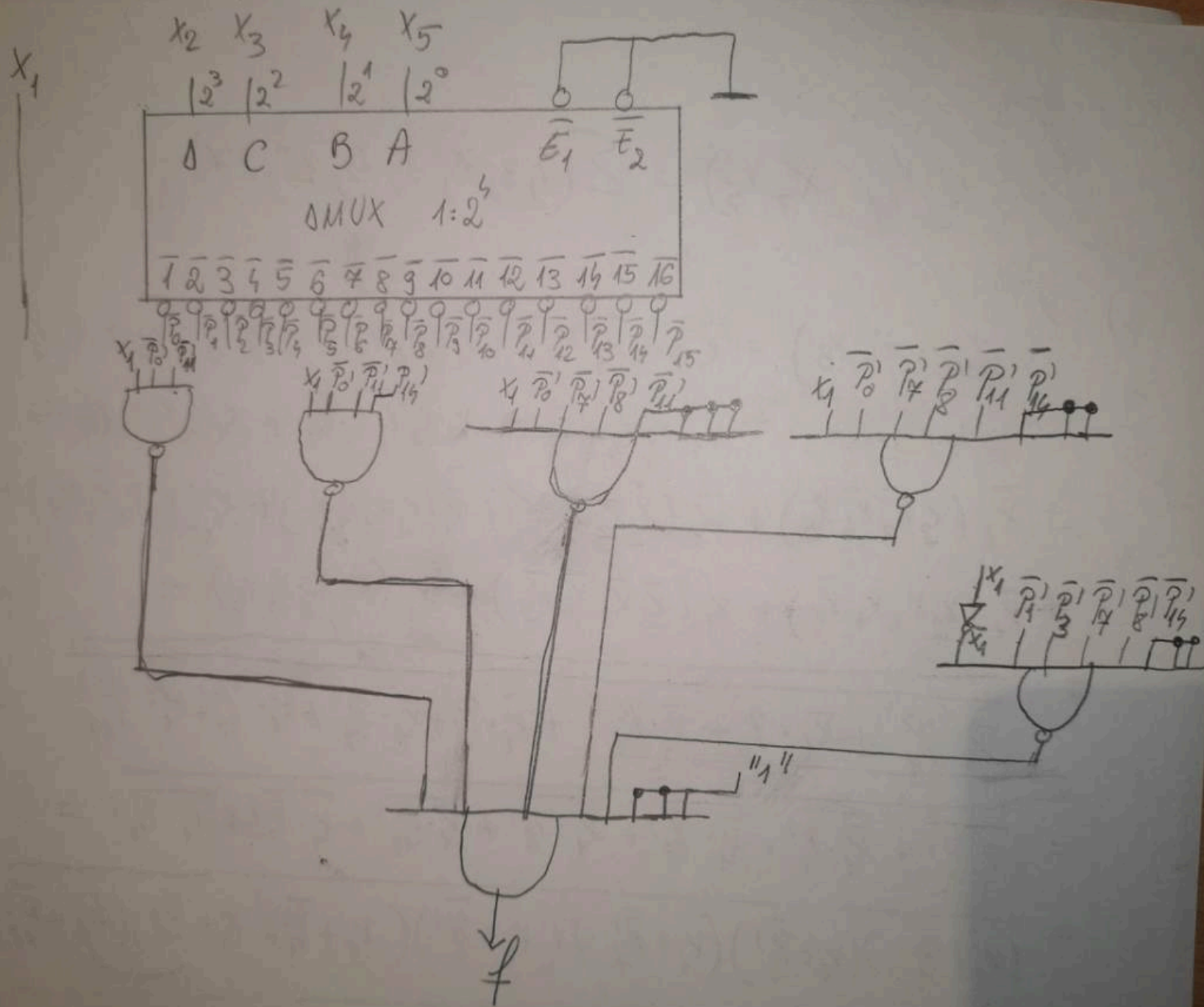
$$= (x_1 \bar{x}_1 + x_1 \bar{p}_0' \bar{p}_{11}' + \bar{x}_1 \bar{p}_1' \bar{p}_3' + \bar{p}_1' \bar{p}_3' \bar{p}_0' \bar{p}_{11}') (x_1 + x_1 \bar{p}_{13}' + x_1 \bar{p}_4' \bar{p}_8' + \bar{p}_4' \bar{p}_8' \bar{p}_{13}') \quad (x \cdot \text{redundant } ab + \bar{a}c + bc = ab + \bar{a}c)$$

$$= (x_1 \bar{p}_0' \bar{p}_{11}' + \bar{x}_1 \bar{p}_1' \bar{p}_3') (x_1 + x_1 \bar{p}_{13}' + x_1 \bar{p}_4' \bar{p}_8' + \bar{p}_4' \bar{p}_8' \bar{p}_{13}') =$$

$$= x_1 \bar{p}_0' \bar{p}_{11}' + x_1 \bar{p}_0' \bar{p}_{11}' \bar{p}_{13}' + x_1 \bar{p}_0' \bar{p}_4' \bar{p}_8' \bar{p}_{11}' + x_1 \bar{p}_0' \bar{p}_4' \bar{p}_8' \bar{p}_{11}' \bar{p}_{13}' + \bar{x}_1 \bar{p}_1' \bar{p}_3' \bar{p}_0' \bar{p}_{11}'$$

$$= \bar{x}_1 \bar{p}_1' \bar{p}_3' \cdot x_1 \bar{p}_0' \bar{p}_{11}' \bar{p}_{13}' \cdot x_1 \bar{p}_0' \bar{p}_4' \bar{p}_8' \bar{p}_{11}' \cdot x_1 \bar{p}_0' \bar{p}_4' \bar{p}_8' \bar{p}_{11}' \bar{p}_{13}' \cdot \bar{x}_1 \bar{p}_1' \bar{p}_3' \bar{p}_0' \bar{p}_{11}'$$

(2)

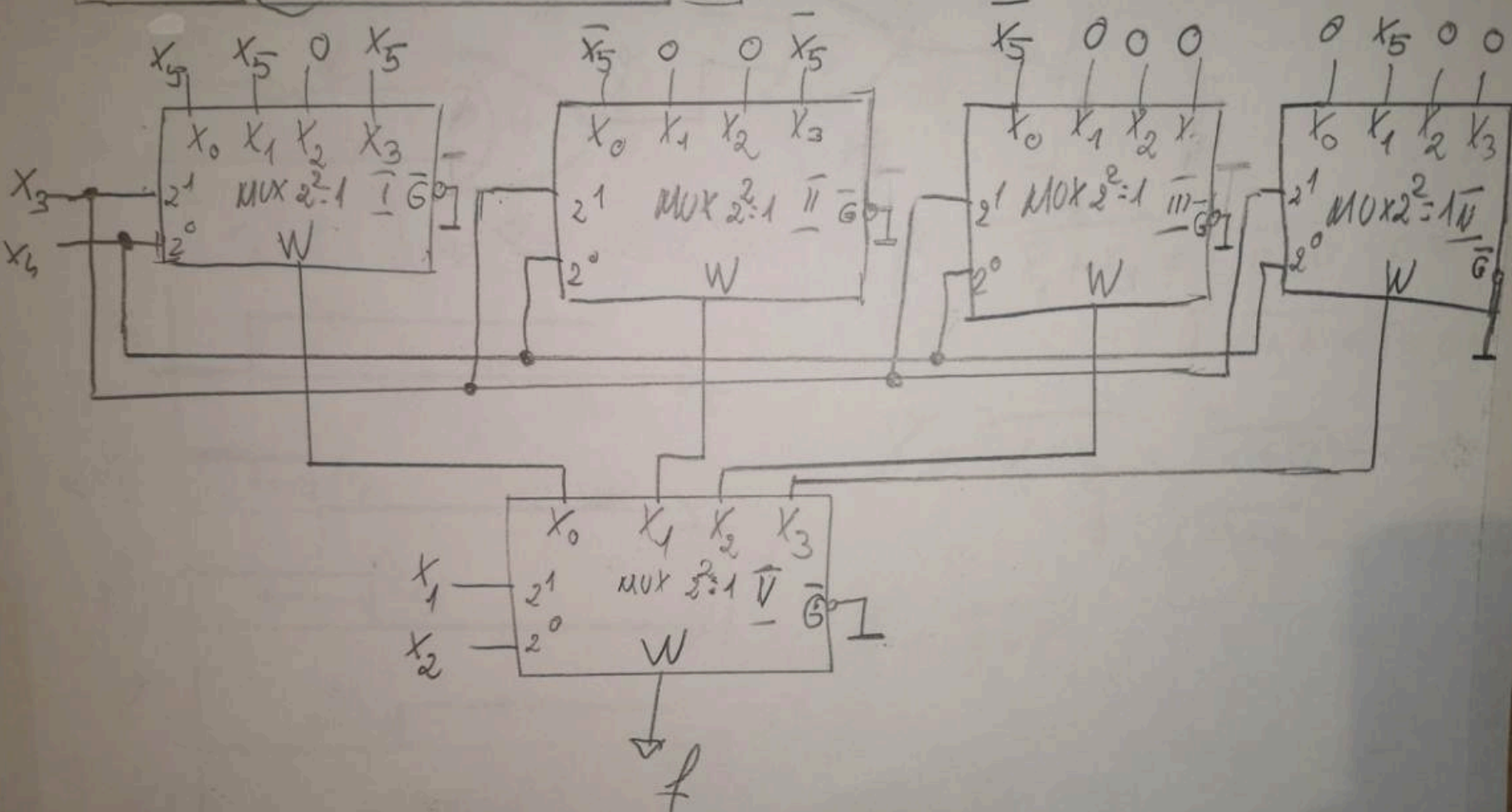


c)

No. CRT	x_1	x_2	x_3	x_4	x_5	f
0	0	0	0	0	0	0
1	0	0	0	0	1	1
2	0	0	0	1	0	0
3	0	0	0	1	1	1
4	0	0	1	0	0	0
5	0	0	1	0	1	0
6	0	0	1	1	0	0
7	0	0	1	1	1	1
8	0	1	0	0	0	1
9	0	1	0	0	1	0
10	0	1	0	1	0	0
11	0	1	0	1	1	0
12	0	1	1	0	0	0
13	0	1	1	0	1	0
14	0	1	1	1	0	1
15	0	1	1	1	1	0

16	1	0	0	0	0	1
17	1	0	0	0	1	0
18	1	0	0	1	0	0
19	1	0	0	1	1	0
20	1	0	1	0	0	0
21	1	0	1	0	1	0
22	1	0	1	1	0	0
23	1	0	1	1	1	0
24	1	1	0	0	0	0
25	1	1	0	0	1	0
26	1	1	0	1	0	0
27	1	1	0	1	1	1
28	1	1	1	0	0	0
29	1	1	1	0	1	0
30	1	1	1	1	0	0
31	1	1	1	1	1	0

15	0	1	1	1	1	0	5	3	1	1	1	1	1	0
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3.

$$x_1 x_2 x_3 \\ 100 \rightarrow 101$$

2)

	x_1	x_2	x_3	f
0	0	0	0	1
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	0
7	1	1	1	1

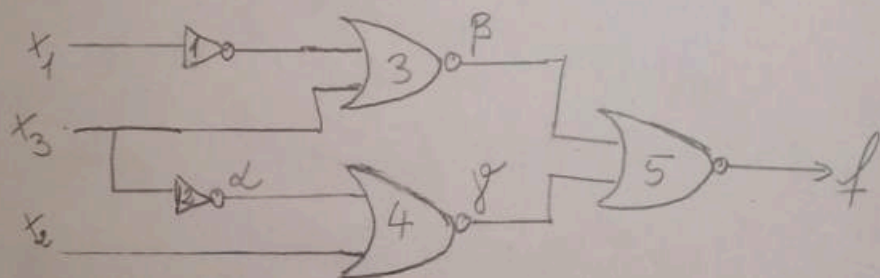
$$- FCD \\ f(x_1, x_2, x_3) = \sum(1, 4, 5, 6) = \bar{x}_1 \bar{x}_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3$$

$$x_1 x_2$$

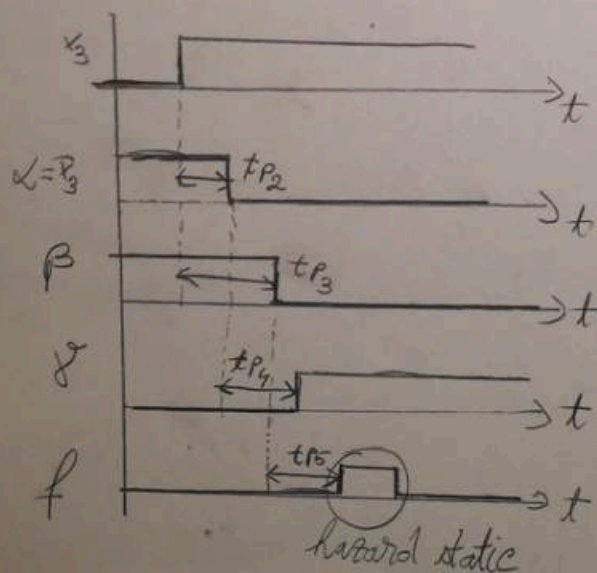
x_3	0	1
0	1	0
1	0	1

$$- FMD \\ f(x_1, x_2, x_3) = x_1 \bar{x}_3 + \bar{x}_2 x_3 ; \quad (\text{va apărea hazardul de carec})$$

$$FMC \\ f(x_1, x_2, x_3) = \overline{(x_1 + x_3)(x_2 + x_3)} = \bar{x}_1 + x_3 + \bar{x}_2 + x_3 ;$$



$$x_1 = 1 \quad x_2 = 0$$



$$\beta = \overline{x_1 + x_3} = \begin{cases} \overline{0+0} = \bar{0} = 1 \\ \overline{0+1} = \bar{1} = 0 \end{cases}$$

$$\gamma = \overline{x_2 + x_3} = \begin{cases} \overline{1+0} = \bar{1} = 0 \\ \overline{0+0} = \bar{0} = 1 \end{cases}$$

$$f = \overline{\beta + \gamma} = \begin{cases} \overline{1+0} = \bar{1} = 0 \\ \overline{0+0} = \bar{0} = 1 \\ \overline{0+1} = \bar{1} = 0 \end{cases}$$

hazard static

b) $f(x_1, x_2, x_3) = x_1 \bar{x}_3 + \bar{x}_2 x_3 + \bar{x}_1 \bar{x}_2$

Se va introduce variabila $x_1 \bar{x}_2$ pentru eliminarea hazardului

$$f^{FMC} = \overline{\bar{x}_1 + x_3} + \overline{x_2 + \bar{x}_3} + \overline{\bar{x}_1 + x_2}$$

Eliminare hazard:

